MATH70097 – Supervised Learning: Assessment 2

March 10, 2026

Deadline: Tuesday 31 March 2026 at 09:00 (London time, BST).

This assignment must be attempted individually; your submission must be your own, unaided work. Candidates are prohibited from discussing assessed coursework and must abide by Imperial College's rules regarding academic integrity and plagiarism. Unless specifically authorized within the assignment instructions, the submission of output from generative AI tools (e.g., ChatGPT) for assessed coursework is prohibited. Violations will be treated as an examination offense. Enabling other candidates to plagiarise your work constitutes an examination offense. To ensure quality assurance is maintained, departments may invite a random selection of students to an 'authenticity interview' on their submitted assessments.

1 Problem Description

You are working with a biomedical research consortium studying how gene expression patterns relate to a continuous biological response (for example, drug sensitivity or protein abundance). The consortium collected measurements from multiple laboratories around the world. Each laboratory performed experiments on different samples using its own equipment and protocols.

For every sample you are given:

- measurements $X \in \mathbb{R}^{40}$ of 40 gene expression features,
- a continuous response variable $y \in \mathbb{R}$ representing the biological outcome of interest.

Because laboratories use slightly different equipment and procedures, measurements coming from the same laboratory may share batch effect.

2 Data overview

The training dataset contains measurements from 40 laboratories. Each laboratory contributed 100 samples. The training dataset includes:

- `y`: response variable,
- `group_id`: the laboratory identifier
- `X1,...,X40`: gene expression features

The test dataset contains new samples from laboratories not present in the training data.

3 Objective

Your goal is to build a regression model that predicts the response y from the gene expression features. Your model should perform well on samples from laboratories that were not seen during training. The target metric is the mean-squared error.

4 Tasks

Write a 4-page report that covers the following points. The report should clearly explain your reasoning and justify the modelling choices you make.

1. **Explore the training data (EDA).**
   Explore relationships between predictors and response, and identify patterns or structure in the dataset. Use appropriate plots or summary statistics to support your analysis.

2. **Model development and interpretation.**
   Train and compare **two** models from the methods covered in the course and compare their predictive performance. Explain your preprocessing steps, modelling choices, and hyperparameters. Interpret the behaviour of your models by discussing whether some particular genes appear more important and what patterns in the data are captured by the models. You should use appropriate tools to support your interpretation.

3. **Validation and model selection.**
   Use an appropriate validation strategy to choose hyperparameters and select your final model. Clearly describe and justify the validation procedure you used. Report an estimate of the generalization error of your final model.

4. **Test prediction.**
   Use your final model to generate predictions for the test dataset. Submissions are evaluated using **mean-squared error** on the test set (lower is better).

4.1 Deliverables

You must submit **three files**.

1. **Report (PDF)**
   A 4-page report. You may write your report in a Jupyter notebook and export it as PDF.
   Name the file: `cid-name_surname-assessment-2.pdf`.

2. **Code (Jupyter notebook, `.ipynb`)**
   The notebook used to produce the results in your report, containing only code required to reproduce the results submitted in the report (figures, tables, and numbers).
   The notebook should run from top to bottom without manual changes.
   Name the file: `cid-name_surname-assessment-2.ipynb`.

3. **Predictions (CSV)**
   A file named `cid-name_surname-assessment-2.csv` with **one row per observation in `test_public.csv`**, containing **exactly two columns**:

   - `obs_id`: the identifier from `test_public.csv`
   - `y_hat`: your **predicted value** for $y \in \mathbb{R}$

Example format:

```
obs_id,y_hat
1,25.78
2,12.81
3,15.40
...
```

5 Marking scheme (100 marks)

Your mark is based on the quality of the analysis, the soundness of the modelling and validation approach, the clarity and reproducibility of your report and code, and the predictive performance of your final model.

The criteria below describe the main aspects considered during grading. They are guidelines rather than a checklist.

5.1 EDA and problem understanding (20 marks)

We assess your understanding of the dataset and the prediction task. Strong submissions:

- clearly describe the structure of the dataset,
- use appropriate plots or summary statistics to explore relationships between variables,
- identify important patterns or potential challenges in the data,
- connect exploratory findings to modelling decisions.

Exploration should be focused and informative, not a collection of unstructured plots.

5.2 Model development and interpretation (25 marks)

We assess how you construct, justify, and interpret predictive models.

You should:

- implement two methods covered in the course,
- explain your preprocessing, modelling choices, and hyperparameters,
- compare the behaviour of different approaches
- interpret the behaviour of your models by discussing which predictors appear influential and what patterns in the data are captured by the models.

Marks are awarded for sound reasoning and clear justification, not simply for trying many models.

5.3 Validation and model selection (25 marks)

We assess whether models are evaluated in a statistically sound way. Strong submissions:

- use an appropriate validation strategy,
- correctly estimate predictive performance during model development,
- clearly justify the chosen validation procedure,
- select the final model based on validation results.

5.4 Interpretation, presentation, and reproducibility (20 points)

We assess the overall quality and clarity of your analysis, including how well you interpret your results and communicate your findings.

Strong submissions:

- clearly explain what worked well and what did not, and provide plausible explanations grounded in the data and the methods used,
- interpret model behaviour (for example, identifying influential variables or patterns captured by the model),
- reflect on limitations of the analysis and suggest possible improvements,
- present results in a clear, well-structured, and concise report.

Code should be organised, commented, and reproducible. The report should be easy to follow and focused on the main findings. Including excessive plots, models, or long unfiltered output will not increase your mark and may reduce clarity.

Marks in this category reflect the overall quality, clarity, and professionalism of the submission

5.5 Performance on held-out test data (10 points)

Predictive performance on the held-out test data contributes up to 10 marks. Lower mean-squared error receives higher marks.